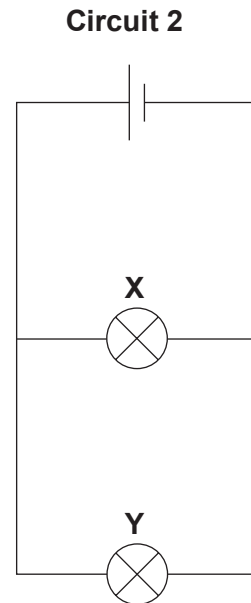
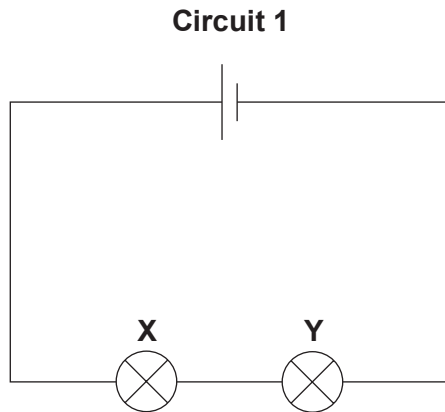


2. A student is studying the properties of different electrical circuits. **X** and **Y** are identical lamps. The student built the following circuits.



- (a) Complete the sentences using the terms in the box.
The terms can be used once, more than once or not at all.

series	less than	parallel	the same as
greater than	simple	current	voltage

- (i) The lamps in **circuit 1** are connected in
 The current through lamp **X** isthe current through lamp **Y**.
 [2]
- (ii) The lamps in **circuit 2** are connected in
 The current through lamp **X** isthe current through lamp **Y**.
 [2]

(b) In **circuit 1** the voltage supplied is 9 V. Each lamp has a resistance of 3 Ω.

(i) Use the equation:

$$\text{current} = \frac{\text{voltage}}{\text{resistance}}$$

to calculate the current in this circuit.

[3]

Current = A

(ii) State what would happen to the current in **circuit 1** if lamp **Y** is removed and the circuit was built with only lamp **X**. [1]

.....

(c) In **circuit 2** the voltage supplied is also 9 V and each bulb has a current of 3 A.

Use the equation

$$\text{power} = \text{voltage} \times \text{current}$$

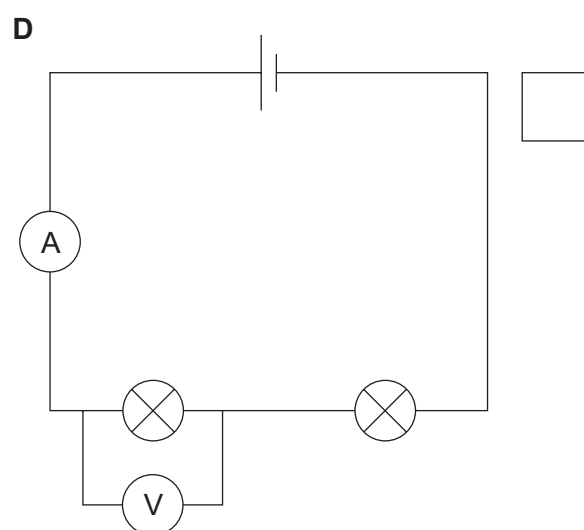
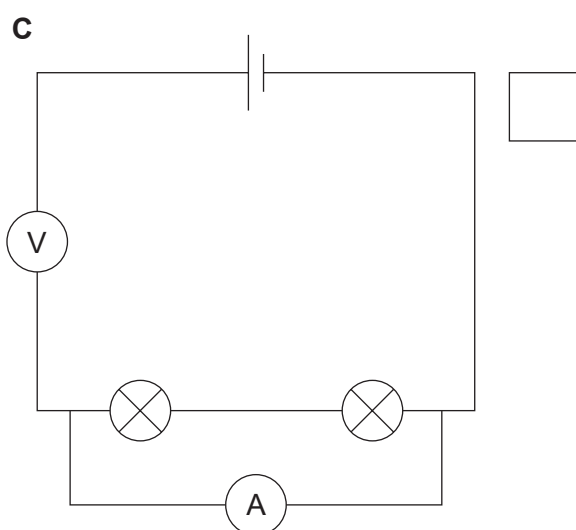
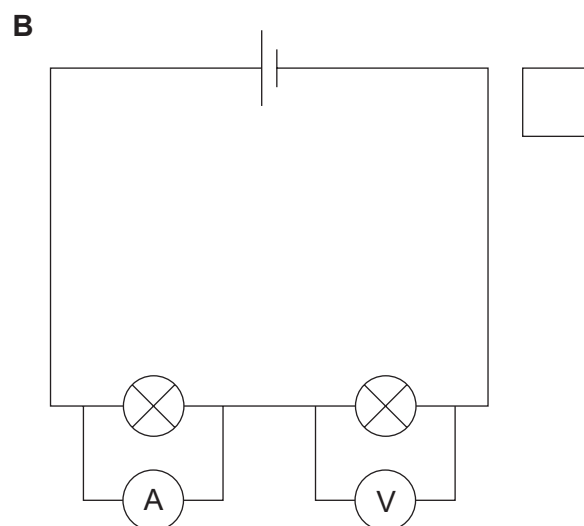
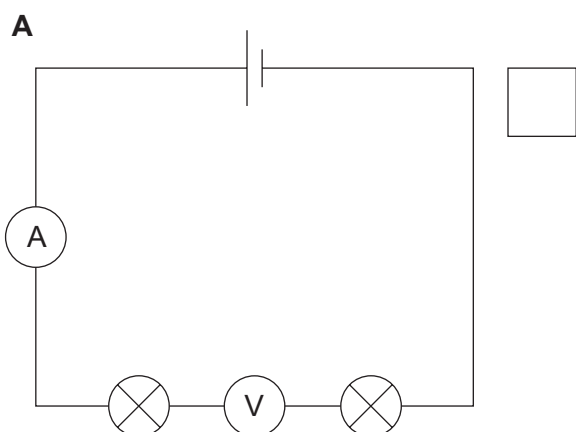
to calculate the power of lamp **X**.

[2]

Power = W

(d) The student set up a circuit to measure the current and voltage correctly.

Use a tick (✓) to identify the circuit diagram that should have been used to measure the voltage. [1]



(e) State **two** reasons why lamps are connected in parallel rather than in series in the home. [2]

1.
2.

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